



V93000 Specifications

Pin Scale 5000 digital card

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PS5000 specifications

Use the Pin Scale 5000 digital card to execute digital tests on EXA Scale™ test systems.

Scope of specifications

Note

The specifications are subject to change without notice. Make sure to refer always to the latest revision found in the most recent SmarTest documentation (System Reference > Specifications).

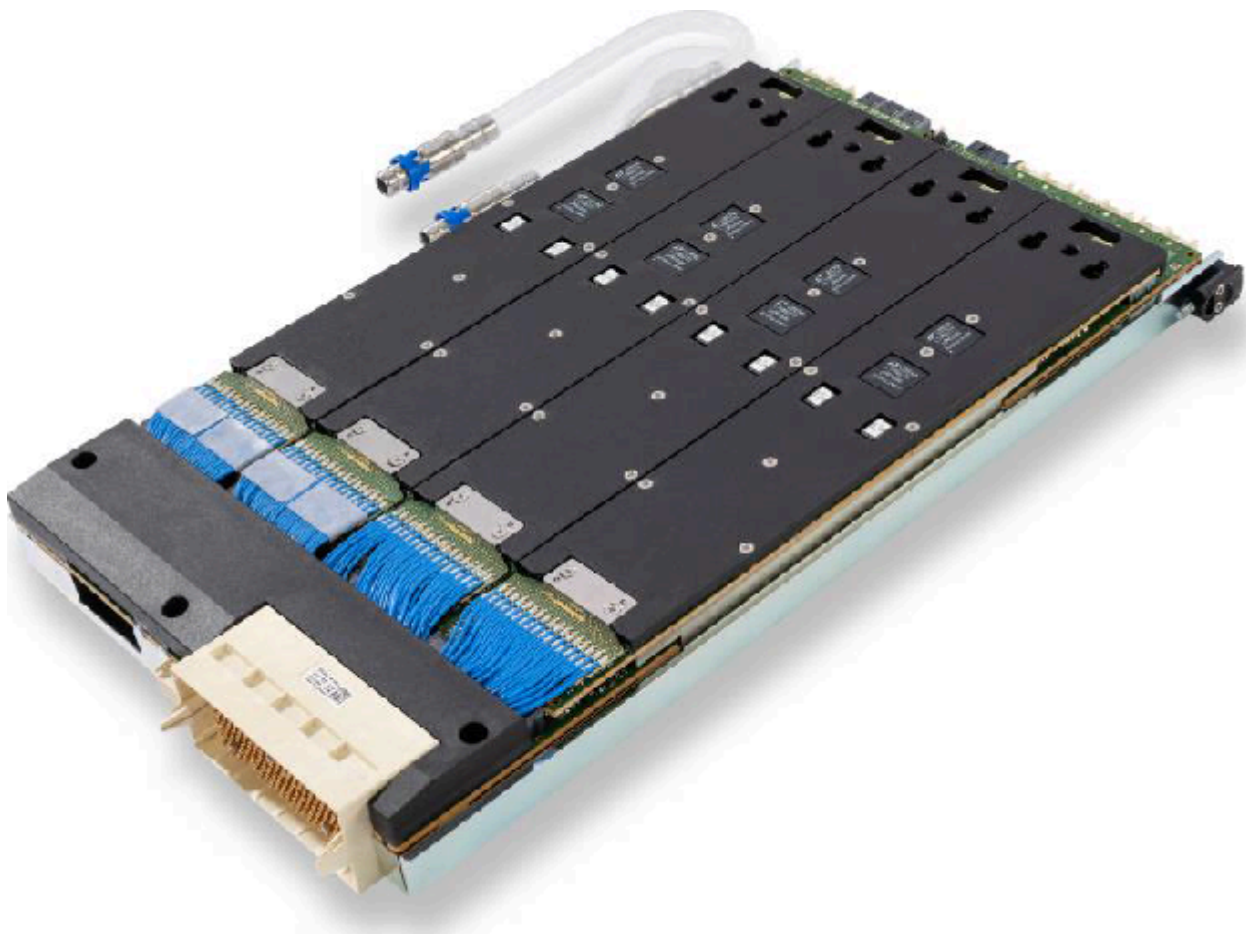
This specification covers the Pin Scale 5000, digital card (part number EK601).

Advantest specifies and verifies the specifications at the DUT interface pogo level. Appropriate DUT board design and layout including proper usage of the GND sense inputs and fixture delay measurements (TDR) is required to ensure specified performance at the device under test.

Specifications define the warranted technical performance covered by products within the stated adjustment interval and operating conditions. Typical specifications define the technical performance met by at least 80 % of all products during product qualification. A typical specification is not covered by product warranty and does not include measurement uncertainties.

General

The Pin Scale 5000 universal digital card is an EXA Scale™ card to execute tests on digital circuitry of a DUT. It is a successor of the Pin Scale 1600 card of Smart Scale™ test systems.



Pin Scale 5000 digital card

Key differences to PS1600

- 256 instead of 128 digital channels
- Pin Scale 5000 can be equipped in EXA Scale™ test systems only
- 1,792 Mi vectors compared to 896 Mi vectors in vector storage per pin, in terms of physical memory per card this is 448 Mibyte instead of 224 Mibyte vector memory per pin
- up to 5000 Mbps instead of 1600 Mbps maximum data rate

Further information

For more information about the Advantest V93000 SOC Series, please visit www.advantest.com.

Contact information

For more information about the Advantest V93000 SOC cards, please contact your local Advantest sales representative.

This information is subject to change without notice.

Pin Scale 5000 basic parameters

Digital channels

Channels	
Maximum number of digital channels	256 per card
Maximum number of pseudo-random binary sequences (PRBSs) per channel	$2^{31} - 1$, programmable seed and compare User defined compliance and serial standard pattern space (64 KiByte per channel)
SmartLoop™	Active, re-timed, differential and single ended, loop direction changeable

Speed

Speed (verification condition: half the noted voltage into 50 Ω)	
Maximum input/output data rate (pattern, PRBS, SmartLoop™)	> 4.0 Gbps but ≤ 5.0 Gbps at ≤ 1.0 V (differential operation only, differential swing ≤ 1.0 V, terminated with 100 Ω, focusing functional test execution) 4.0 Gbps at ≤ 1.2 V 2.1 Gbps at ≤ 3.0 V 1 Gbps at ≤ 7.0 V
Maximum clock rate	> 2.0 GHz ≤ 2.5 GHz at ≤ 1.0 V (differential operation only, differential swing ≤ 1.0 V, terminated with 100 Ω, focusing functional test execution) 2 GHz at ≤ 1.2 V 1.05 GHz at ≤ 3.0 V 500 MHz at ≤ 7.0 V

Operation and maintenance

For the definition of week, month, year in the context of a maintenance period refer to *Units of maintenance periods*.

Environment, operation, and maintenance	
Warm-up time	60 min
External adjustment period	6 months
Internal adjustment period, valid at ambient temperature within ±2.5 # of internal adjustment (auto calibration) temperature	3 months
Operating temperature range	+15 # to +30 # (+59 # to +86 #)
Temperature range for warranted specifications	+20 # to +30 # (+68 # to +86 #)
Maximum humidity at +30 # (+86 #)	< 70% relative humidity, non-condensing

Environment, operation, and maintenance	
Storage temperature range, without the presence of water	-40 # to +70 # (-40 # to +158 #)

Pin Scale 5000 vector memory

General

The maximum amount of memory to be used for storing vectors is 448 MiByte per pin. The usable amount of memory depends on the given memory license. For maximum vector memory license, vector memory will be shared with a configurable amount of memory for sequencer instructions, result buffers, and level/timing sets.

Vector memory content can be shared across pins within a group of 8 pins for input, output. Result and capture memory for an output and input/output pin will still be available per individual pin. With current SmarTest software, sharing of memory content across pins is only supported for the same pin across sites.

Functional vector modes

Different x-modes for functional vectors and scan vectors can be mixed within a pattern burst.

Functional vector mode	Maximum data rate	Device cycles per tester cycle	Maximum timing edges per device cycle or vector	Vectors per byte	Maximum vectors per pin
X1-mode	1 Gbps	1	8 drive + 8 receive + 8 tristate	1	448 M
X2-mode	2 Gbps	2	4 drive + 4 receive + 4 tristate	2	896 M
X4-mode	4 Gbps	4	2 drive + 2 receive + 2 tristate	4	1,792 M
X5...X8-mode	5 Gbps	5...8	1 drive + 1 receive + 1 tristate	5...8	2,240...3,584 M

Scan vector modes

Every pin can be used as scan-in or scan-out pin. Switching between scan-in and scan-out within a pattern burst is supported. Scan vector mode means that due to the reduced states set, the vectors can be stored with better efficiency using either X8 (scan-in) or X5 (scan-out) mode.

Scan vector mode	Maximum data rate	States per scan vector	Maximum scan vectors per pin	Maximum scan vectors per 8 pins	Maximum scan vectors per card
Scan-in	5 Gbps	0 / 1	3,584 M	28,672 M	917,504 M
Scan-out	5 Gbps	H / L / X	2,240 M	17,920 M	573,440 M

Pin Scale 5000 timing

Accuracy

It is $t_{EPA} = t_{OTA}/2$. For a graphical representation refer to *Pin Scale 5000 terms* on page 27.

OTA and EPA

OTA and EPA		
Test system	OTA	±120 ps
	EPA	±60 ps
Per card	OTA	±80 ps
	EPA	±40 ps
Per channel pair	OTA	±64 ps
	EPA	±32 ps
Edge placement resolution		1 ps
Edge placement range		-4 to +32 sequencer periods
Clock jitter	2.0 GHz	350 fs rms (typical) ^[1]
	1.0 GHz	400 fs rms (typical) ^[1]
	156.25 MHz	700 fs rms (typical) ^[2]
	100 MHz	1200 fs rms (typical) ^[2]

Vector period

Vector period	
Vector period range, the change of period on the fly is supported across the full period range	0.2 ns (5 device cycles per tester cycle, x-mode) to 100,000 ns
Vector period accuracy	±15 ppm of period setting
Vector period resolution	0.001 fs

Eye height

Eye height is measured with the following figures:

- $V_{swing} = 500$ mV into 50 Ω at a single connection (single-ended signaling)
- $V_{swing} = 1000$ mV into 100 Ω at a pair of legs (differential signaling).

For a graphical representation refer to *Pin Scale 5000 terms* on page 27.

^[1] Measurement conditions: 800 mV differential swing into 100 Ω . Fast driver is being used.
Measurement: Keysight E5052B, phase noise integration range 1 kHz ... 40 MHz

^[2] Measurement conditions: 800 mV differential swing into 100 Ω . Fast driver is being used.
Measurement: Keysight E5052B, phase noise integration range 12 kHz ... 20 MHz

Eye height at maximum data rates ^[3]		
Single-ended	$\Delta V_{\text{SignalEyeHeight}}$ of driver	> 40% of V_{swing} (typical)
	$\Delta V_{\text{SignalEyeHeight}}$ of driver/receiver pair	> 25% of V_{swing}
Differential	$\Delta V_{\text{SignalEyeHeight}}$ of driver	> 40% of V_{swing} (typical) > 25% at > 4.0 Gbps
	$\Delta V_{\text{SignalEyeHeight}}$ of driver/receiver pair	> 25% of V_{swing} > 10% at > 4.0 Gbps

Transition tracking

Transition tracking			
Tracking range	at 4 Gbps	with jitter frequency = 100 kHz	± 128 user intervals (UI)
	at 4 Gbps	with jitter frequency = 100 MHz	± 0.6 user intervals (UI)
Tracking frequency	at 4 Gbps	with jitter amplitude = 1 UI p-p	150 MHz
	at 4 Gbps	with jitter amplitude = 2 UI p-p	37.5 MHz
	at 1 Gbps	with jitter amplitude = 1 UI p-p	75 MHz

Drive jitter injection

Drive jitter injection	
Timing jitter range	± 128 UI at 100 kHz
	± 0.6 UI at 100 MHz
Jitter frequency range	100 kHz to 100 MHz

^[3] Single ended operation specified up to 4Gbps (see Speed on page 6). Test pattern: PRBS7, starting from clock pattern (non-static state), verification at pogo., excluding influence of performance verification board.

Pin Scale 5000 driver

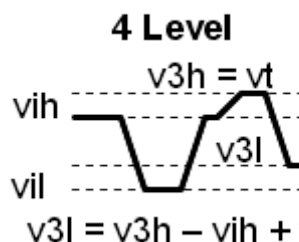
AC performance

AC performance (verification condition: half the noted voltage, terminated into 50 Ω)			
Maximum transition time (typical)	Fast mode 20% to 80%	at 1.2 V swing	130 ps
	Standard mode 10% to 90%	at 1.8 V swing	450 ps
		at 3.0 V swing	750 ps

DC performance

The measurements are done with the default equalization settings (0).

DC performance high, low, 3rd and 4th level, specified into open			
High level (V_{ih}) range at $ V_{GS} \leq 100$ mV	Fast mode		-1.5 V to +3.5 V ^[4]
	Standard mode		-1.5 V to +7.0 V
3rd level (V_{3h} ^[5]) range at $ V_{GS} \leq 100$ mV	Fast mode		-1.5 V to +3.5 V ^[4]
	Standard mode		-1.5 V to +6.5 V
Low level (V_{il} , V_{3l}) range at $ V_{GS} \leq 100$ mV	Fast mode		-1.5 V to +3.5 V ^[4]
	Standard mode		-1.5 V to +6.5 V
Level swing ($V_{ih} - V_{il} = V_{3h} - V_{3l}$) range	Fast mode		20 mV to 1.2 V
	Standard mode		50 mV to 8.0 V
Level resolution			1 mV
Level accuracy at $V_{GS} = 0$ V	Standard mode		± 5 mV
	Fast mode 2-level mode	High voltage V_{ih}	± 20 mV
		Swing $V_{swing} = V_{ih} - V_{il}$	± 10 mV
		Differential swing	± 20 mV
	Fast mode 4-level mode	High voltage V_{ih} , V_{3h}	± 25 mV
		Swing $V_{swing} = V_{ih} - V_{il}$	± 15 mV
		Differential swing	± 22 mV
		Shifted swing $V_{3h} - V_{3l}$ ^[6]	± 1 mV $\pm 1\% * (V_{3h} - V_{ih})$



^[4] See also Pin Scale 5000 ground sense input on page 23.

^[5] Depending on the application, V_{3h} is also used as termination voltage V_t .

^[6] The shifted swing is the change of swing transitioning between V_{ih} and V_{3h} .

Driver impedance

Driver impedance, including pogo cable assembly		
Source impedance	$-1.5\text{ V} \leq V_{il}, V_{ih}, V_{3l}, V_{3h}, V_t \leq 3.0\text{ V}$	$50\ \Omega \pm 3\ \Omega$
	$3.0\text{ V} < V_{il}, V_{ih}, V_{3h}, V_{3l}, V_t$	50 Ω (typical)

Pin Scale 5000 receiver

AC performance

AC performance	
Intrinsic transition time 20% to 80%	144 ps (typical)

DC performance (single ended)

The measurements are done with the default equalization settings (0).

DC performance (single ended)		
Input voltage and threshold range at $ V_{GS} \leq 100$ mV		-1.5 V to +6.5 V
Threshold voltage resolution		1 mV
Threshold voltage accuracy at $V_{GS} = 0$ mV		± 5 mV
Maximum input leakage current	High-Z	± 2 μ A
	DCL off	$\pm (10 \text{ nA} + 1 \text{ nA/V} * V_{act})$ ^[7]

DC performance (differential)

The measurements are done with the default equalization settings (0).

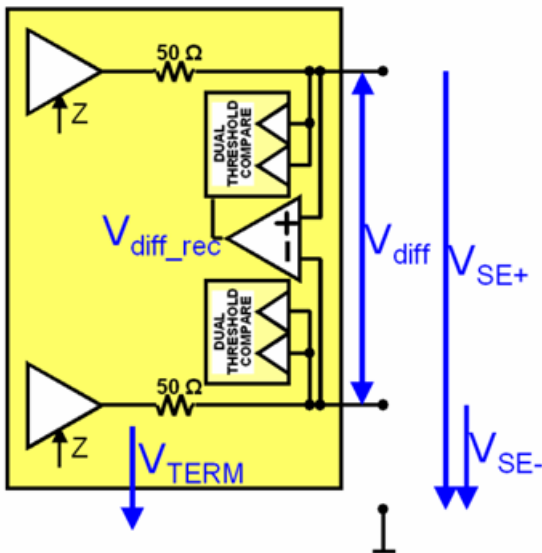
DC performance (differential)		
Input voltage range at $ V_{GS} \leq 100$ mV		-1.5 V to +6.5 V
Maximum differential input voltage		± 8 V
Differential threshold voltage range		-1.0 V to +1.0 V
Threshold voltage resolution		1 mV
Threshold voltage accuracy		± 5 mV
Maximum input leakage current	High-Z	± 2 μ A
	DCL off	$\pm (10 \text{ nA} + 1 \text{ nA/V} * V_{act})$

Termination

Termination		
Single-ended impedance	$-1.5 \text{ V} \leq V_{SE} \leq 3.0 \text{ V}$	$50 \Omega \pm 3 \Omega$
	$V_{SE} > 3.0 \text{ V}$	50Ω (typical)
Center tap differential impedance	$-1.5 \text{ V} \leq V_{SE} \leq 3.0 \text{ V}$	$100 \Omega \pm 6 \Omega$
	$V_{SE} > 3.0 \text{ V}$	100Ω (typical)
Termination voltage range at $ V_{GS} \leq 100$ mV		-1.5 V to +6.5 V
Termination voltage resolution		1 mV
Termination voltage accuracy		± 5 mV

V_{SE} is the single-ended, resulting voltage at the pin line. See the figure below.

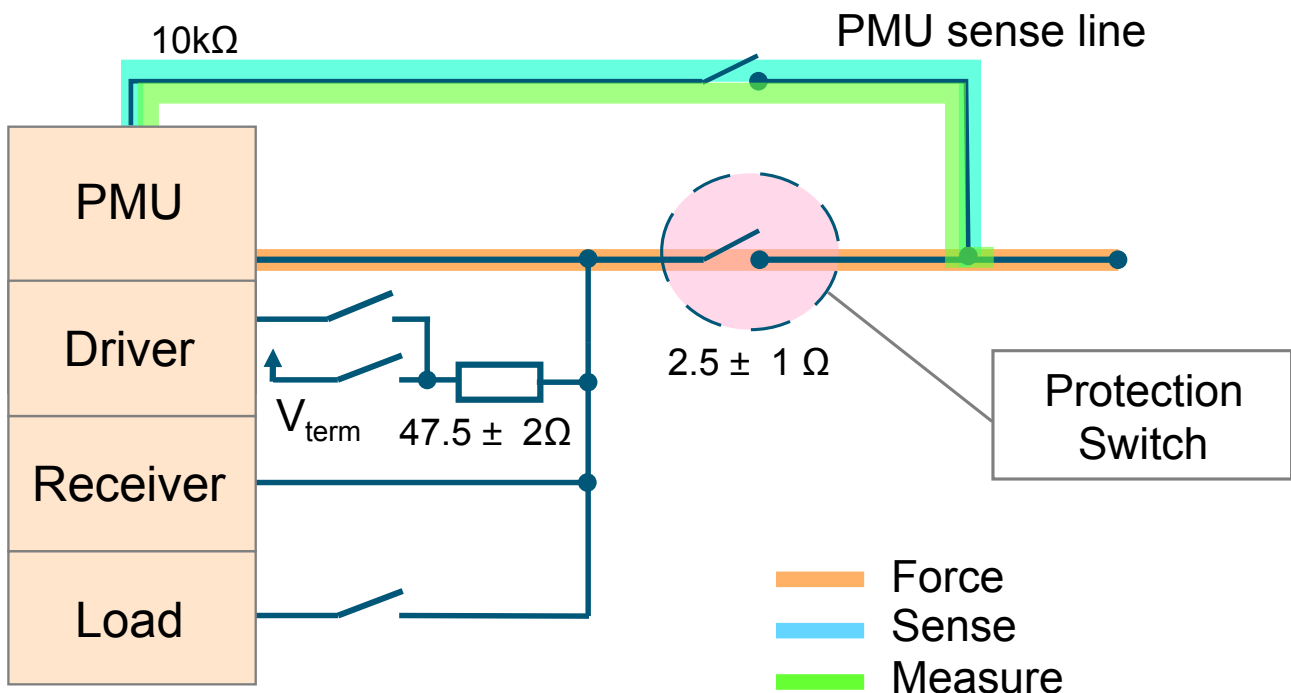
^[7] V_{act} is the actual voltage of the channel, here the external voltage forced by the DUT.



Pin Scale 5000 voltage definitions and termination conditions

Application note: Terminated voltage measurements using the receiver

When the comparators of the receiver circuit are used to measure voltages take into consideration that the protection switch has a 2.5Ω series resistance. This is part of the 50Ω termination the DUT sees but contributes to the measurement results with an I_R drop in case the measurement is performed in loaded / terminated conditions as shown below.



PS5000 receiver: Simplified diagram of the pin electronics

The term loaded conditions here denotes the usage of DCL (drive, compare, load circuitry) termination, either driver or active load termination. When the PPMU is involved, this I_R drop is not contributing because of the sense/measure point being placed in front of the protection switch.

The term $I_A \cdot R$ in the *voltage measure specification*, relates to wiring and contact resistance caused I_R drop. It is important to understand that the protection resistance discussed here comes on top as term only relevant for digital receivers.

Pin Scale 5000 time measurement unit

DC performance

Refer to the data given in *Pin Scale 5000 receiver* on page 13.

AC performance

AC performance	
Maximum input frequency	2.1 GHz
Linearity error	± 4 ps
Accuracy, period or frequency measurement	± 2 ps ^[8]
Accuracy, single timestamp to ARM (t_0)	± 30 ps ^[9]
Accuracy, skew / propagation delay (different pins)	$\pm 2 * (t_{EPA} + 30 \text{ ps})$ ^[10]
Intrinsic jitter	1 ps rms (typical)
Resolution	1.75 ps
Maximum sampling rate	375 Msps

^[8] Requires an averaging time of 500 periods. In a measurement of a multiple events in TMU measurements, periods can be determined very accurately. For validation, the measurement is stimulated with the driver. The specified value refers to the variation of events, equally spaced by the stimulus pattern, from the expected result.

^[9] Measurement of a single event in a pattern, stimulated with the driver. The driver EPA uncertainty has been removed from the result.

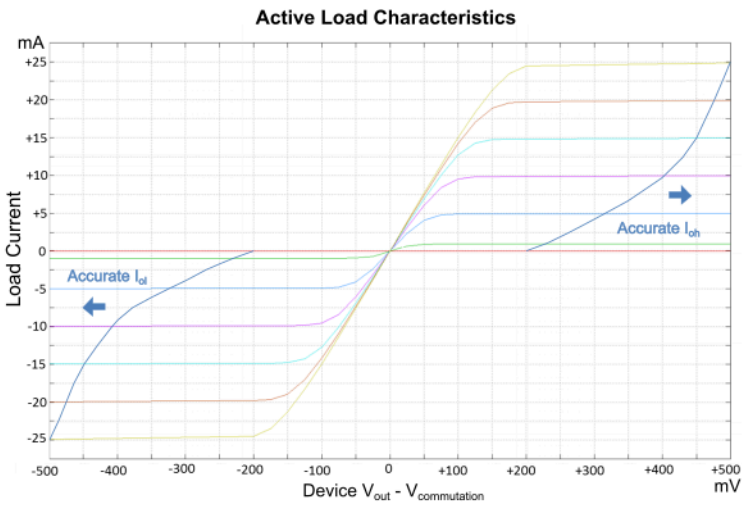
^[10] This accuracy number represents the skew between measurements of 2 different channels.

Pin Scale 5000 programmable load

The programmable load is not available for fast mode.

Programmable load		
Load current range (I_{OH} , I_{OL})		0 mA to 25 mA
Current resolution		12.5 μ A
Current accuracy		$\pm 75 \mu$ A $\pm 0.5\%$
Commutation voltage range (V_{COM}) at $ V_{GS} \leq 100$ mV	I_{OH} , $I_{OL} < 1$ mA	-1.5 V to +6.5 V
	I_{OH} , $I_{OL} \geq 1$ mA	-1.0 V to +5.5 V
Commutation voltage resolution		1 mV
Commutation voltage accuracy		± 100 mV

Active load



Pin Scale 5000 active load characteristics

Pin Scale 5000 programmable clamp

The programmable clamp is not available for fast mode.

Programmable clamp	
Clamp voltage range at $ V_{GS} \leq 100 \text{ mV}$	-2.0 V to +7.0 V
Maximum clamp voltage window	9 V
Clamp voltage resolution	1 mV
Clamp voltage accuracy, verified with a measurement current of 900 μA	$\pm 100 \text{ mV}$

The sustainable total load may not exceed 5,000 mA per card.

Pin Scale 5000 parametric measurement unit

To achieve the specified accuracy, the user must take into account the number of internal samples given in the next sections for the averaging. The *averages* property of the instrument specifies how many times these internal samples are executed to obtain the final measured value. Total number of samples = number of internal samples as defined in the tables below * *averages*. The default value for *averages* is 1.

Voltage force and measure

I_a is the actual current. R is the maximum wiring resistance of 1 Ω .

Voltage force and measure		
Voltage force / measure range at $ V_{GS} \leq 100$ mV	$I \leq 25$ mA	-1.50 V to +6.45 V
	$I > 25$ mA	-0.60 V to +5.60 V
Voltage force	Resolution	200 μ V
	Accuracy with force sensing at $V_{GS} = 0$ V	± 2 mV
	Accuracy without force sensing at $V_{GS} = 0$ V	$\pm (2 \text{ mV} + I_a * R)$
Voltage measure	Accuracy ADC through PPMU	
	Resolution	50 μ V
	Accuracy	
	measure (-0.2 V to +4.0 V) at $V_{GS} = 0$ V, number of internal samples for averaging 16 ^[11]	$\pm (2 \text{ mV} + I_a * R)$
	measure (-1.5 V to +6.5 V) at $V_{GS} = 0$ V, number of internal samples for averaging 16 ^[11]	$\pm (4 \text{ mV} + I_a * R)$
Voltage measure	Accuracy pass/fail and SAR value through PPMU level comparators	
	Resolution	1 mV
	Accuracy	
	compare (-1.5 V to +6.5 V) at $V_{GS} = 0$ V	$\pm (10 \text{ mV} + I_a * R)$
	measure (-1.5 V to +6.5 V) at $V_{GS} = 0$ V	$\pm (5 \text{ mV} + I_a * R) \pm 0.5\%$ of reading
Capacitive loading	≤ 10 nF	

^[11] For compatibility and correlation reasons, SmarTest 8 continues to use an internal averaging factor of 32.

Current force and measure

Current force and measure						
Range	± 120 mA, ganging of both channels of a differential channel pair, this feature will be available in a later SmarTest release	± 60 mA	± 1 mA	± 100 µA	± 10 µA	± 2 µA
Resolution	40 µA	20 µA	500 nA	50 nA	5 nA	1 nA
Measure accuracy ADC	± 100 µA ± 1%, number of internal samples for averaging 4	± 50 µA ± 0.5%, number of internal samples for averaging 4	± 1.25 µA ± 0.5%, number of internal samples for averaging 4	± 200 nA ± 0.5%, number of internal samples for averaging 16	± 50 nA ± 0.5%, number of internal samples for averaging 32	± 10 nA ± 0.5%, number of internal samples for averaging 64
Measure typical accuracy SAR, always one sample	± 400 µA ± 1.0%	± 200 µA ± 0.5%	± 20 µA ± 0.5%	± 20 µA	± 2 µA	± 0.5 µA
Force accuracy	± 100 µA ± 0.4%	± 50 µA ± 0.2%	± 5 µA ± 0.5%	± 500 nA ± 0.5%	± 100 nA ± 0.5%	± 40 nA ± 0.5%

Voltage clamp

The voltage clamp is available in current force mode only.

Voltage clamp	
Voltage clamp range	-1.5 V to +6.5 V at $ V_{GS} \leq 100$ mV
Clamp resolution	1 mV
Clamp accuracy, verified with a measurement current of 900 µA	± 100 mV at $V_{GS} = 0$ mV

To avoid clipping effects, the clamp must be specified with a sufficient distance to the expected measurement result. In symbolic notation, $|\text{clamp}| > \max(|\text{expected measurement result range}|) + |\text{clamp accuracy}| + |\text{result accuracy}|$.

Current clamp

The current clamp is available in voltage force mode only.

Current clamp						
Current range	± 120 mA, ganging of both channels of a differential channel pair	± 60 mA	± 1 mA	± 100 µA	± 10 µA	± 2 µA
Clamp range	± 36 mA to 132 mA	± 18 mA to 66 mA	± 0.3 mA to 1.1 mA	± 30 µA to 110 µA	± 3 µA to 11 µA	± 0.6 µA to 2.2 µA

Current clamp						
Resolution	40 µA	20 µA	500 nA	50 nA	50 nA	50 nA
Clamp accuracy	± 4 mA	± 2 mA	± 100 µA	± 10 µA	± 1 µA typical	± 0.2 µA typical

To avoid clipping effects, the clamp must be specified with a sufficient distance to the expected measurement result. In symbolic notation, $|clamp| > \max(|expected\ measurement\ result\ range|) + |clamp\ accuracy| + |result\ accuracy|$.

The sustainable total load may not exceed 5000 mA per card.

Pin Scale 5000 precision voltage measure (board ADC)

Precision voltage measure with board ADC

One board ADC (BADC) is shared between 16 channels for *precision* voltage measure.

Precision voltage measure		
Voltage range		-1.5 V to +6.5 V at $ V_{GS} \leq 100$ mV
Measurement resolution		10 μ V
Measurement accuracy	Ground sense connected and operational, or relative measurement without ground sense. ^[12]	± 500 μ V at $V_{GS} = 0$ V, number of internal samples for averaging 16 ^[13]
	Ground sense disconnected or not enabled	± 1000 μ V, number of internal samples for averaging 16 ^[13]
Maximum input leakage current in voltage range -1.5 V to +6.5 V	High-Z	± 2 μ A
	DCL off	$\pm (10 \text{ nA} + 1 \text{ nA/V} \cdot V_{act})$ ^[14]

The accuracy of a relative measurement refers to two consecutive measurements on the same pin or pins sharing the same board ADC, with or without ground sense.

Accuracy of relative (delta) measurements

Voltage range -0.1 V to 5.1 V	Delta < 250mV	± 100 μ V
Voltage range -0.1 V to 5.1 V		± 150 μ V
Voltage range -1.5 V to 6.5 V		± 200 μ V

The minimum number of samples for averaging is 512.

Remarks

This section of the specification is valid for high impedance measurements, which means neither involving the PPMU circuit nor terminating the signal with the DCL (driver/comparator/load) circuitry.

Such measurements can be performed the following ways:

dcVI instrument: Choose `connectMode = highImpedance` to select the precision voltage measure mode.

digInOut instrument: Using the `frontendConnection` action, both the `digitalFrontend` and `dcFrontend` must be set to `disconnected`. Then a `vmeas` action can be invoked by specifying `direct =enabled` and `highAccuracy = true`. In all other scenarios, the PPMU circuit performs the measurement, and PPMU accuracies apply, see *Pin Scale 5000 parametric measurement unit* on page 18.

^[13] For compatibility and correlation reasons, SmarTest 8 continues to use an internal averaging factor of 32.

^[12] Relative measurement: accuracy between two consecutive measurements on same pin or pins sharing the same Board ADC, with or without Ground Sense

^[14] V_{act} is the actual voltage of the channel, here the external voltage forced by the DUT.

Pin Scale 5000 high voltage protection

High voltage protection

High voltage protection	
Compliance range	-2.5 V...+7.5 V
Overvoltage cut-off range	-40 V ... -5.5 V / +10 V ... +80 V
Overvoltage reaction time	≤ 15 ns
Overcurrent cut-off range	≥ 150 mA
Overcurrent reaction time	≤ 500 ns

Pin Scale 5000 ground sense input

Ground sense input

For information about V_{GS} refer to *Pin Scale 5000 terms* on page 27.

Ground sense input		
Ground sense inputs	SD and HD pogo blocks	One ground sense per 16 channels
	VHD pogo blocks	One ground sense per 32 channels
Ground sense voltage range		$ V_{GS} \leq 500$ mV, for $ V_{GS} > 100$ mV voltage ranges specified have to be reduced accordingly

Pin Scale 5000 arbitrary waveform generator (AWG)

Arbitrary waveform generator (AWG)

I_a is the actual current. R is the maximum wiring resistance of 1 Ω .

		PPMU ^[15]
Voltage range, single ended		-1.5 V to +6.5 V at $ V_{GS} \leq 100$ mV ^[16]
Accuracy		$\pm (2 \text{ mV} + I_a * R)$ at $V_{GS} = 0$ mV
Maximum current		± 60 mA
Selectable hardware filters		None
Analog bandwidth (typical)		3 kHz
Sampling rate		0...150 ksps
Waveform memory		16 Msamples ^[17]
Fan out (signal distribution matrix)		N/A
Offset error (typical)		$\pm 360 \mu\text{V}$ at $V_{GS} = 0$ mV
Gain error (typical)		± 65 ppm
INL (typical)		$\pm 400 \mu\text{V}$
DNL (typical)		$\pm 360 \mu\text{V}$
Noise (typical)	22 Hz ... 22 kHz	N/A
	22 Hz ... 100 kHz	400 μV_{RMS}
	1 kHz	83 dB at 7.9 V_{pp} , values ranging from -1.5V ... +6.4V.
SNR (typical)	10 kHz	N/A
	20 kHz	N/A
	1 kHz	-75 dBc at 7.9 V_{pp} , values ranging from -1.5V ... +6.4V.
THD (typical)	10 kHz	N/A
	20 kHz	N/A
	1 kHz	-92 dBc at 7.9 V_{pp} , values ranging from -1.5V ... +6.4V.
SFDR (typical)	10 kHz	N/A
	20 kHz	N/A

^[15] Per Pin PMU used as AWG, available for every pin, to be set up with `dcVI.sourceWaveform` instrument .

^[16] V_{GS} is the voltage at the ground sense input.

^[17] Common memory for vector data, waveform samples and capture results.

Pin Scale 5000 voltage digitizer

Voltage digitizer BADC

Voltage digitizer BADC, per 16 pins		
Voltage range	-1.5 V to +6.5 V at $ V_{GS} \leq 100$ mV	
Resolution	20 bit (10 μ V)	
DC accuracy	± 500 μ V at $V_{GS} = 0$ mV	
Analog bandwidth (typical, -3 dB)	113 kHz	
Sampling rate	0...1 Msps	
Capture memory	16 Msamples ^[18]	
Offset error (typical)	± 150 μ V	
Gain error (typical)	± 35 ppm	
INL (typical)	± 50 μ V	
DNL (typical)	± 16 μ V	
RMS Noise (typical data)	22 Hz ... 130 kHz	65 μ V _{RMS}
THD (up to 5 th harmonic, typical)	1 kHz (100 ksps sampling rate, no filtering), 1 V _p (1 VP = 2 VPP, values in a 0 V ... 2 V range)	- 98 dBc
	1 kHz , 3 V _p (3 VP = 6 VPP, values in a -0.5 V ... + 5.5 V range)	- 103 dBc
	10 kHz (100 ksps sampling rate, no filtering), 1 V _p	- 93 dBc
	10 kHz, 3 V _p	- 87 dBc
	20 kHz (200 ksps sampling rate, no filtering), 1 V _p	92 dBc
	20 kHz , 3 V _p	105 dBc
SFDR ^[19] (typical data)	1 kHz (100 ksps sampling rate, no filtering), 1 V _p	97 dBc
	1 kHz , 3 V _p	105 dBc
	10 kHz (100 ksps sampling rate, no filtering), 1 V _p	97 dBc
	10 kHz, 3 V _p	105 dBc
	20 kHz (200 ksps sampling rate, no filtering), 1 V _p	97 dBc
	20 kHz , 3 V _p	103 dBc
SNR (typical)	1 kHz (100 ksps sampling rate, no filtering), 1 V _p	77 dB
	1 kHz , 3 V _p	87 dB

^[18] Common memory for vector data, waveform samples and capture results.

^[19] SFDR specifies the available dynamic range as the difference in magnitude between the amplitude of the fundamental and the amplitude of the largest spurious component excluding harmonics.

Voltage digitizer BADC, per 16 pins		
	10 kHz (100 ksps sampling rate, no filtering), 1 V _p	77 dB
	10 kHz, 3 V _p	86 dB
	20 kHz (200 ksps sampling rate, no filtering), 1 V _p	77 dB
	20 kHz , 3 V _p	85 dB

Further specification values on measuring waveforms with the board ADC are provided in *dcVI measureWaveform actions*.

Pin Scale 5000 terms

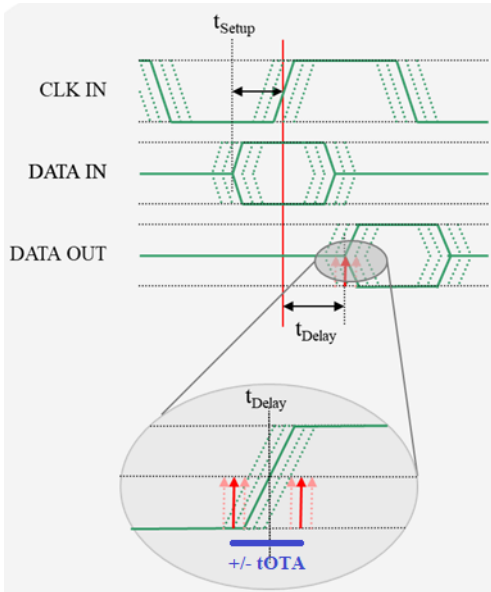
Edge placement accuracy t_{EPA}

This is the worst case timing error of driver edges (or receiver edges) over all channels and settings.

It is $t_{EPA} = t_{OTA}/2$, refer to the explanation given below.

Overall timing accuracy t_{OTA}

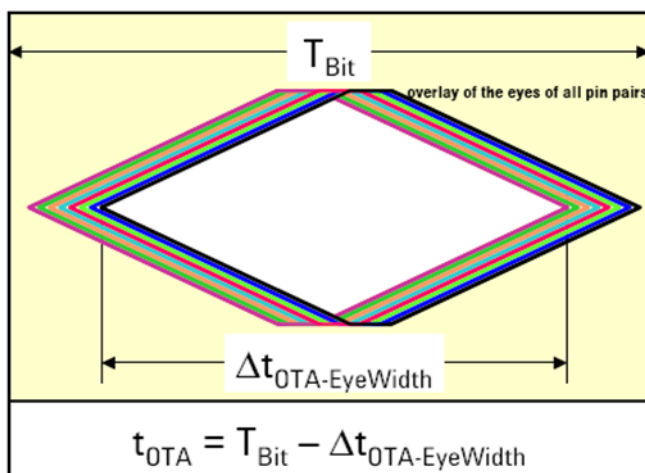
This is the worst case timing error between driver and receiver edges over all channels and settings. In the figure below, $t_{Setup} + t_{Delay}$ is the mean timing value for which a tester edge strobe is programmed. The maximum variation of $(t_{Setup} + t_{Delay})$ is given by $\pm t_{OTA}$.



OTA in edge diagram

It is $t_{OTA} = t_{EPA}$ of the driver + t_{EPA} of the receiver = $2 * t_{EPA}$.

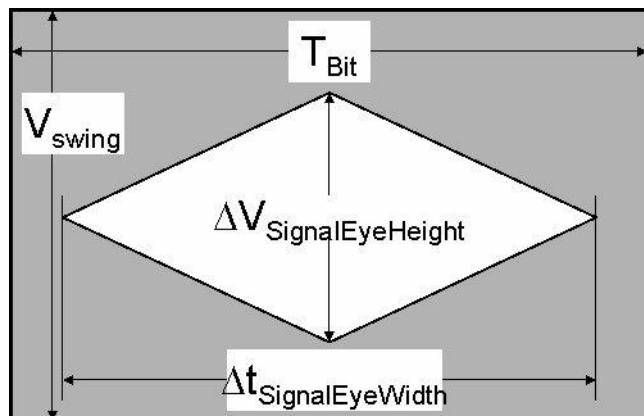
For an illustration of OTA in the eye diagram refer to the figure below.



OTA in eye diagram

Eye height and eye width

For an illustration of the parameters $\Delta V_{\text{SignalEyeHeight}}$ and $\Delta t_{\text{SignalEyeWidth}}$ refer to the figure below.



Eye height and eye width

Ground sense voltage V_{GS}

To support compliance to voltage force and measure specifications at the DUT pin, the PS5000 card provides *ground sense inputs* which, if properly connected to the DUT, enable to compensate for $\geq 97\%$ (or $\geq 94\%$ for the digital driver) of DUT ground offsets caused by test fixture GND wiring / plane resistance.

Linearity error of AC performance

This is the maximum deviation of a periodic time stamp produced by an instrument from its *conventional true value*, given a constant conventional true value of the period between the time stamps. It is caused by non-linear characteristics of the hardware. This error refers to the timing accuracy of the I/O functions specified to be executed by the test system.

Range

This is the interval between the minimum value and the maximum value which have been specified for a *measurand*.

Full scale range (FSR)

The full scale range of a *measurand* is the maximum range of this measurand that can be processed according to its operational specification. If the measurand is found outside the FSR, distorting effects will have to be expected on the measurement result, such as clipping.

Sensitivity

This is the quotient $\Delta y / \Delta x$, with the minimum value of a *measurand* difference Δx which will produce a detectable change Δy on the *result of the measurement* y , $y = f(x)$.

Resolution

This is the minimum value of the *measurand* Δx used in the definition of sensitivity.